

AirWave

The First Hands free, Distance
Aware, Gesture Controlled Smart
Speaker

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Motivation

Motivation

Modern Speakers have Fallen Behind!

- **Hands are busy:** Cooking, lifting, studying - stopping to adjust music breaks flow
- **No Awareness:** Speakers don't know our distance or context, so volume constantly feels wrong
- **Voice isn't Enough:** Noisy shared spaces make commands unreliable and not private



Our Solution: AirWave!

A hands-free, distance-aware speaker using simple gestures + voice commands for seamless music control.



Our Product Goals

AirWave

Hands free, distance aware, gesture controlled speaker.

Key Features

- Wearable IMU control for intuitive, motion-based playback
- Voice command integration for seamless, hands-free operation
- Face-tracking camera that dynamically adjusts volume based on user distance
- Emotion-aware recommendations that suggest playlists to match the listener's mood
- LED feedback system providing clear visual confirmation of recognized gestures



Overall Design

Layout

1) Raspberry Pi 1

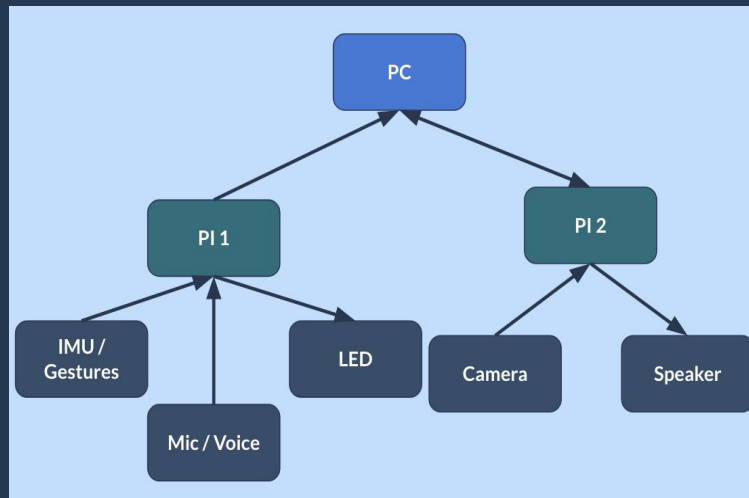
Handles microphone input, LED control, and wearable IMU data. Processes voice commands and gesture-based controls locally

2) Laptop / Host Computer

Operates as the MQTT broker for system communication. Runs the user dashboard for manual controls, monitoring, and personalization.

3) Raspberry Pi 2

Manages speaker output and camera input. Performs user tracking and emotion detection to enable adaptive responses.



Pi 1

1) IMU Gesture Control

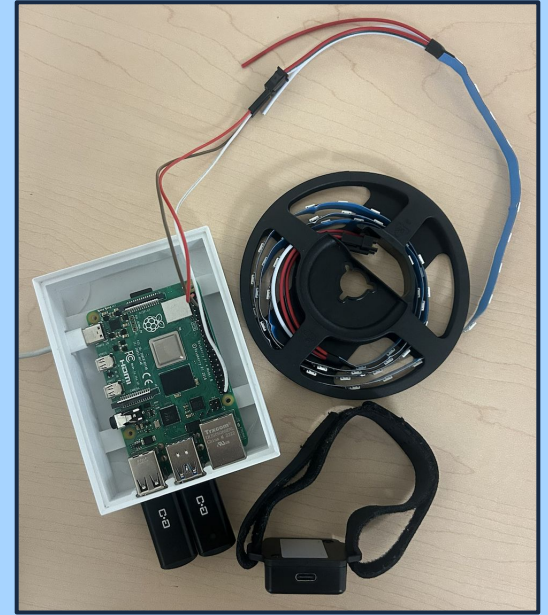
- A wearable IMU bracelet streams motion data to Pi via Bluetooth
- A double-flick gesture activates Command Mode for intentional input
- In Idle Mode, the user can move naturally without triggering actions
- Supported gestures: Play, Pause, Next Track, Previous Track

2) Voice Commands

- Audio captured by the microphone is processed locally on Raspberry Pi 1
- Supported commands: Play, Pause, Next/Previous Track, Volume Up/Down

3) LED Feedback

- Visual LED confirmation indicates successful command recognition
- Light Dancing



IMU - Gesture Recognition

- IMU bracelet connected to PI 1 via Bluetooth
- Gestures we trained using a calibration script and recordings from multiple positions
- Recognized gestures are published to the MQTT Broker then received by PI 2 to actually control playback
- Published commands are displayed on user dashboard for real-time monitoring



Voice Commands - Speech Processing

- Microphone captures user speech on Raspberry Pi 1
- Audio is processed locally using offline speech recognition
- Supported commands: Play, Pause, Next/Previous Track, Volume Up/Down
- Recognized commands are published to the MQTT broker
- Commands are received by Pi 2 to control playback in real time



LED Feedback Implementation

- WS2812B LED strip connected to Raspberry Pi 1 GPIO
- Command confirmation flash
 - Green flash → Gesture recognized
 - Blue flash → Voice command recognized
- Music-reactive “light dancing” synchronized with playback
- Provides clear visual confirmation and enhances user experience



Pi 2

1) User Tracking & Distance Awareness

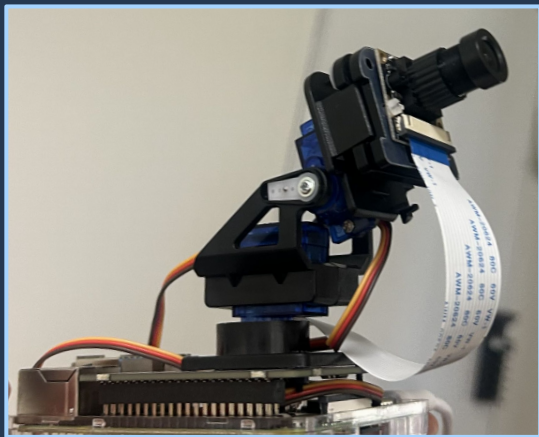
- Raspberry Pi 2 processes live video to detect and track the user's face and position
- Distance estimation is used to dynamically adjust speaker volume for a consistent listening experience

2) Audio Output

- The speaker connects to Raspberry Pi 2 via aux for playback
- Users stream their music by connecting to the Pi over Bluetooth

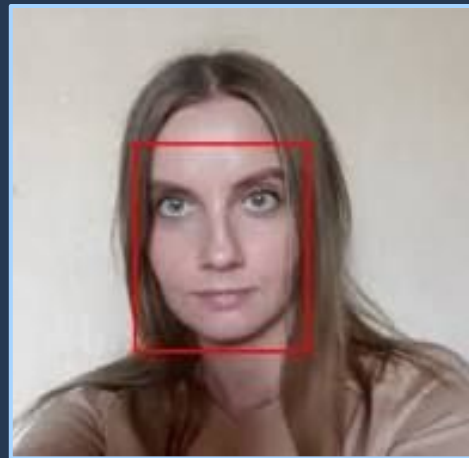
3) Emotion Detection

- Facial analysis classifies user mood (Happy, Sad, or Chill)
- Detected emotion is published to the user dashboard
- The dashboard provides a direct link to a recommended playlist tailored to the user's mood



User Tracking Implementation

- Live video is processed in real time to detect and lock onto the user's face using OpenCV cascade classifiers
- A pan/tilt servo mount continuously adjusts the camera angle to keep the user centered in frame
- Bounding box area is used to estimate distance, triggering automatic speaker volume adjustments as the user moves closer or farther away



Emotion Detection Implementation

- OpenCV measures smile presence, eye count, brightness, contrast, and skin warmth - each factor is weighted and scored to determine the dominant mood
- Analyzes the user's face to classify their mood as Happy, Sad, or Calm
- Mood is detected periodically every 15 seconds while the user is being tracked
- A matching playlist is recommended based on the detected mood and pushed directly to the dashboard



User Setup - Spotify Integration

Audio playback is controlled via Spotify → users must connect their account to Air Wave

- Users are directed to a setup screen on launch
- Connects to AirWave via Mac hostname and Spotify login
- Guided setup walks users through configuration
- Automatically launches and connects Raspberry Pis



 **AirWave**
Welcome! Let's get you set up.

1 MQTT Broker Configuration

On your Mac, run:
`echo "$(scutil --get LocalHostName).local"`

MQTT Broker Address:

2 Spotify Configuration (Required)

1. Go to: developer.spotify.com/dashboard
2. Click "Log In" (top right)
⚠️ **LOGIN with the Spotify account you'll use to play music**
3. Click your profile → Dashboard → "Create App"
4. App Settings:
 - App Name: AirWave (or anything)
 - Redirect URI: `http://127.0.0.1:3888/callback`
5. Copy your Client ID and Client Secret below

Note: Spotify Premium is required for AirWave to work.

Spotify Client ID:

Spotify Client Secret:

3 Connect Your Phone to Pi 2

■ **IMPORTANT:** BEFORE clicking "Start AirWave" below, pair your phone:

1. Open Bluetooth settings on your phone
2. Look for **"PiSpeaker"**
3. Tap to connect
4. This is where your music will play!

 **Start AirWave**

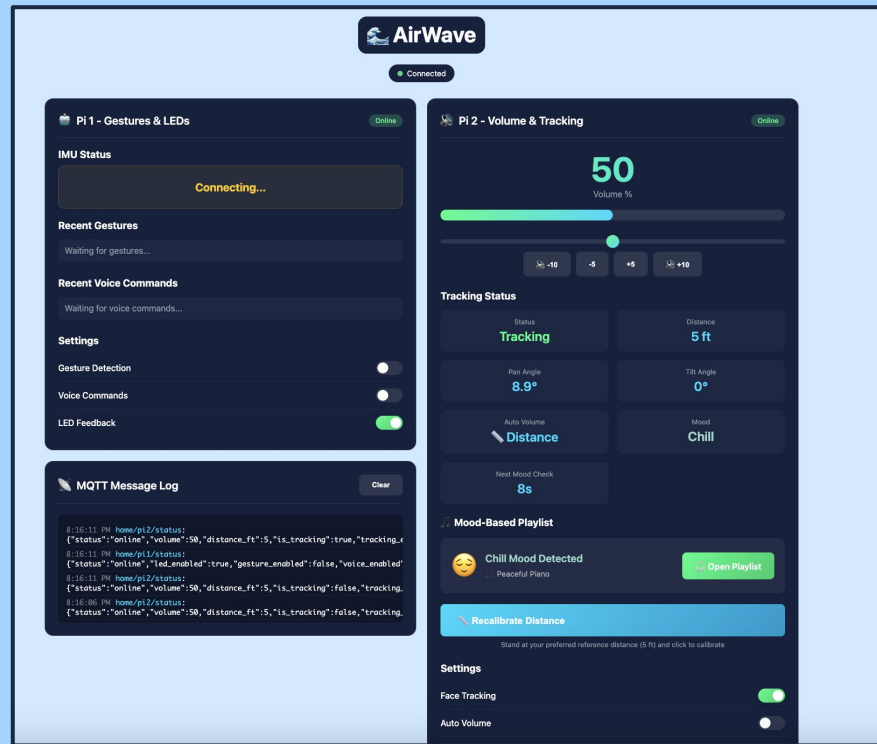
User Interface - Dashboard

Pi 1 Controls

- Connection status indicator showing when both Raspberry Pis are online
- Displays current IMU state (Idle or Command Mode)
- User controls to enable/disable:
 - Gesture detection
 - Voice commands
 - Auto volume adjustment
- Live command log showing recently recognized gestures and voice inputs

Pi 2 Controls

- Option to override automatic distance-based volume with manual adjustment
- Real-time camera tracking status display
- Shows most recently detected mood with a link to a recommended playlist



Differentiating Factors

Gesture Control

- IMU-based gesture controls allow voiceless users to interact with the system without speech commands.

Low-Latency Voice Detection

- Uses offline voice detection, eliminating cloud processing delays and enabling faster system response.

User-Friendly Setup & Interface

- A guided dashboard interface abstracts system complexity, allowing users to set up and operate AirWave without needing technical knowledge.

Personalized User Controls

- The dashboard enables users to customize which interaction features are active.

Testing

Gesture and Voice Testing

1) Gestures

Each gesture was performed N times by multiple users from varied starting positions to validate robustness.

2) Voice

Each voice command was repeated N times across different users, voices, and pitch ranges to ensure reliable recognition.

Metrics Evaluated

- Accuracy – Correct recognition rate
- Latency – Time to process and respond
- False-Trigger Rate – Frequency of unintended activations



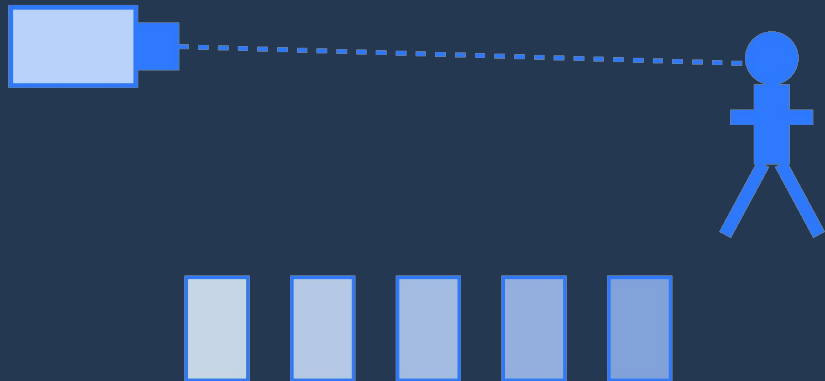
Camera Tracking and Distance Control

What we test: Person detection, Continuous tracking

Protocol: subject moves at varying speeds and distances

Metrics:

- Tracking Stability
- Positional Accuracy
- Max Tracking Speed
- Volume smoothness

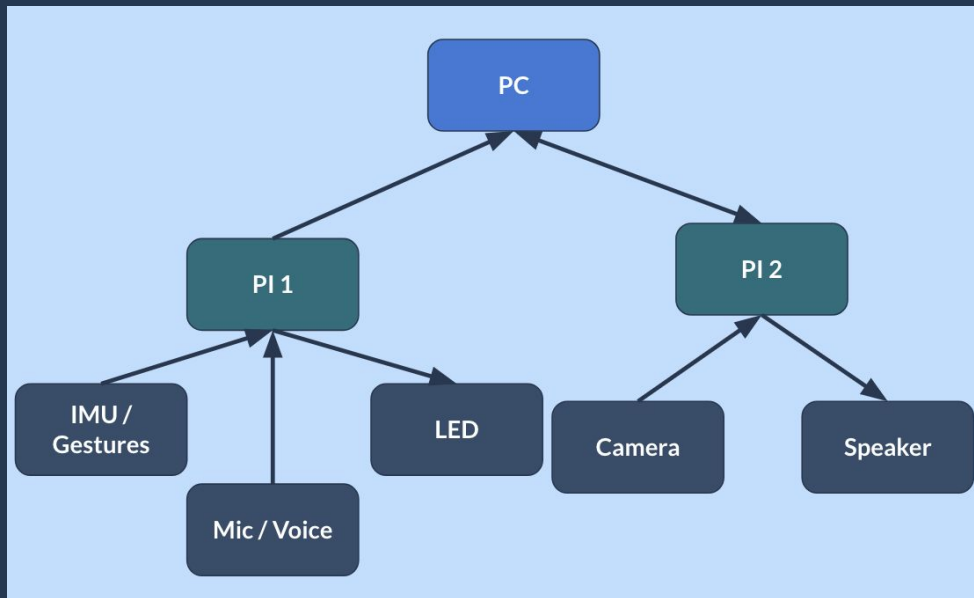


Integrated System Testing

Test Scenarios

Metrics:

- End-to-End Latency
- User Satisfaction
- System uptime/stability



User-Testing Feedback

Improved Gesture Controls

- Made IMU gestures more reliable and tied them directly to the user's playback session
- Added clearer directional guidance on the dashboard

Enhanced Visual Feedback

- Added LED "light dancing" effects alongside gesture confirmation feedback

More Reliable Voice Commands

- Improved voice command recognition with clear prompts indicating when the user should speak
- Reduced microphone sensitivity to prevent music playback from triggering commands

Improved User Accessibility

- Added clear step-by-step instructions on the dashboard to guide user interactions

Limitations

- Camera tracking latency
- Mic sensitivity
- Volume latency

Responsibilities

Leah

- IMU gesture recognition
- Spotify audio playback
- User dashboard
- Launcher.py script
- Integration

Matthew

- Distance based volume control
- Bluetooth integration
- Mood detection
- Camera Tracking

Annique

- Voice recognition/commands
- Light dancing and LED feedback
- Testing

Drew

- Distance based volume control
- Camera tracking
- User dashboard
- Integration
- Testing

Timeline

Fall Quarter

Week 8

- Wrote the foundational code for IMU gesture and voice recognition

Week 9

- Wrote the foundation code for the camera for distance tracking and volume adjustment
- Tested individual components

Week 10

- Develop a preliminary draft of the distance-controlled volume adjustment code
- Integrate basic gesture controls and voice recognition into single script the design

Winter Quarter

Week 1-3

- Implemented **emotion detection** and synchronize **LED feedback**
- Implement full integration
- Connected audio playback to spotify

Week 4-5

- Performed **comprehensive functional testing** to verify that each feature operates as intended.
- Created **launcher script** that makes user setup easy

Week 6-9

- Continued full integration testing
- Did **extensive user testing** and handle any **edge cases**

Week 10

- Continued testing
- Final Presentation

Future Applications

Future Applications

1) Party Mode

- Enable multiple users to interact with the camera and control the speaker simultaneously.

2) Improved Volume Feedback

- Reduce latency between user distance detection and volume adjustment.

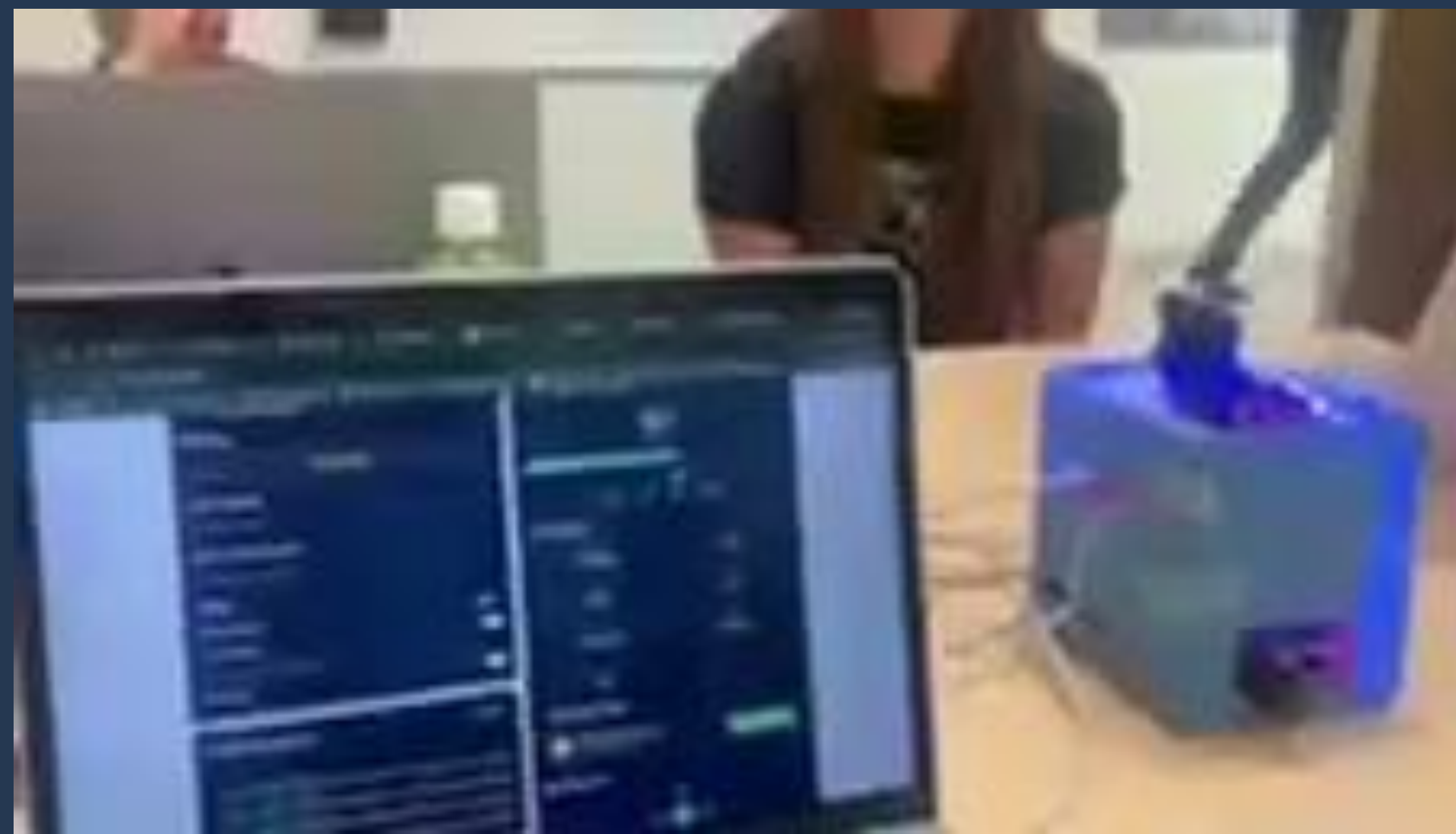
3) Multi-User Gesture Tracking

- Integrate a second Bluetooth IMU to support interactions from multiple users.

4) Expanded Music Platform Support

- Allow connections with Apple Music in addition to Spotify (current playback is limited to Spotify Premium).

Demo



Thank you!